Correlation between Monthly Figures for Traffic Accidents and Weather Figures in Munich Region

Main Question

How do weather conditions affect traffic accidents? Is the number of accidents increasing or decreasing compared to the previous year?

Description

Hundreds of accidents have been reported in recent months in various parts of Germany. This project aims to identify how the rates of accidents have been affected by weather conditions in Munich. Specifically, it seeks to establish a correlation between weather conditions and road accidents.

Data Sources

Datasource 1: Monthly Figures for Traffic Accidents

- **Metadata URL**: Monatszahlen Verkehrsunfälle
- **Data URL**: Accident Data CSV
- **Data Type**: CSV
- **Content**: Monthly figures for traffic accidents including the number of accidents, fatalities, and injuries.

Datasource 2: Monthly Weather Figures

- **Metadata URL**: Monatszahlen Witterung
- **Data URL**: Weather Data CSV
- **Data Type**: CSV
- **Content**: Monthly weather statistics including temperature, precipitation, and other relevant weather conditions.

Data Structure and Quality

Both datasets are provided in CSV format, ensuring ease of integration and processing. The traffic accident data includes date, number of accidents, injuries, and fatalities, while the weather data includes monthly statistics on temperature, precipitation, and other weather conditions. The data quality is generally good, with no significant missing values, allowing for reliable analysis.

Data Licenses

The datasets are sourced from open data platforms under standard open-data licenses, allowing for use in this project.

Data Pipeline

High-Level Overview

1. **Data Extraction**: Automated downloading of CSV files from the provided URLs.
2. **Data Cleaning**: Handling missing values, correcting inconsistencies, and ensuring uniform data formats.
3. **Data Transformation**: Aggregating monthly data, normalizing different units, and merging datasets based on date.
4. **Analysis**: Calculating correlations and generating visualizations.
5. **Error Handling**: Implementing checks for data integrity and fallback mechanisms for missing data.

Technologies Used

- **Python**: Primary language for scripting and data manipulation.
- **Pandas**: Library for data processing and analysis.
- **Matplotlib/Seaborn**: Libraries for data visualization.
- **Jupyter Notebook**: Environment for development and documentation.

Transformation and Cleaning Steps

- **Parsing Dates**: Ensuring all dates are in a consistent format.
- **Handling Missing Values**: Filling or interpolating missing weather data points.
- **Normalization**: Standardizing units of measurement for weather data.
- **Merging Datasets**: Combining accident and weather data on a monthly basis.

Problems Encountered and Solutions

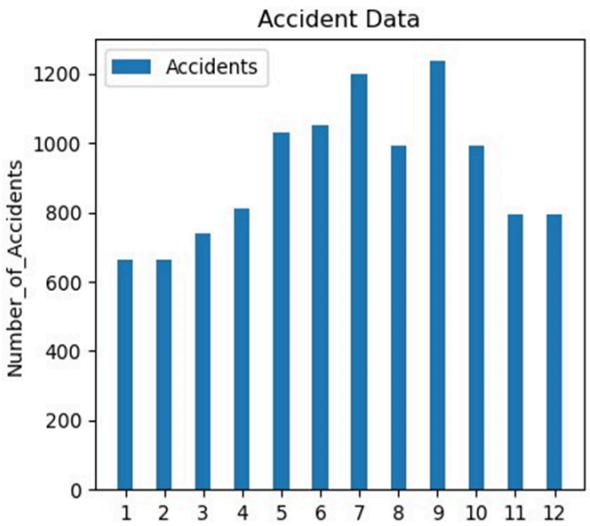
- **Missing Data**: Addressed by interpolating missing weather data points.
- **Inconsistent Formats**: Standardized date and unit formats across datasets.
- **Data Merging Issues**: Ensured precise matching of records based on month and year.

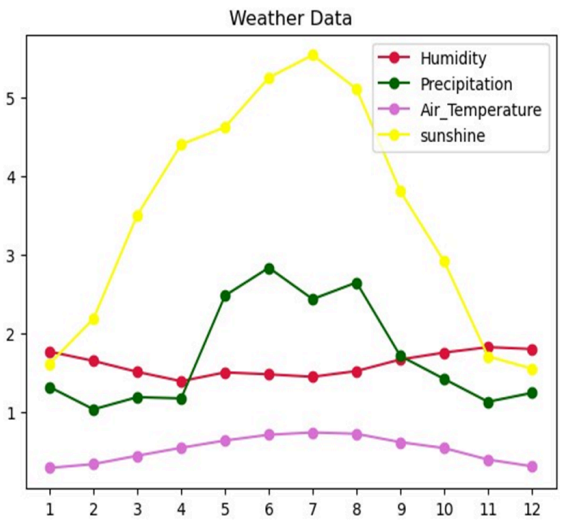
Error Handling

- **Data Integrity Checks**: Automated validation of data quality before processing.
- **Fallback Mechanisms**: Using previous month's data for missing records to ensure continuity.

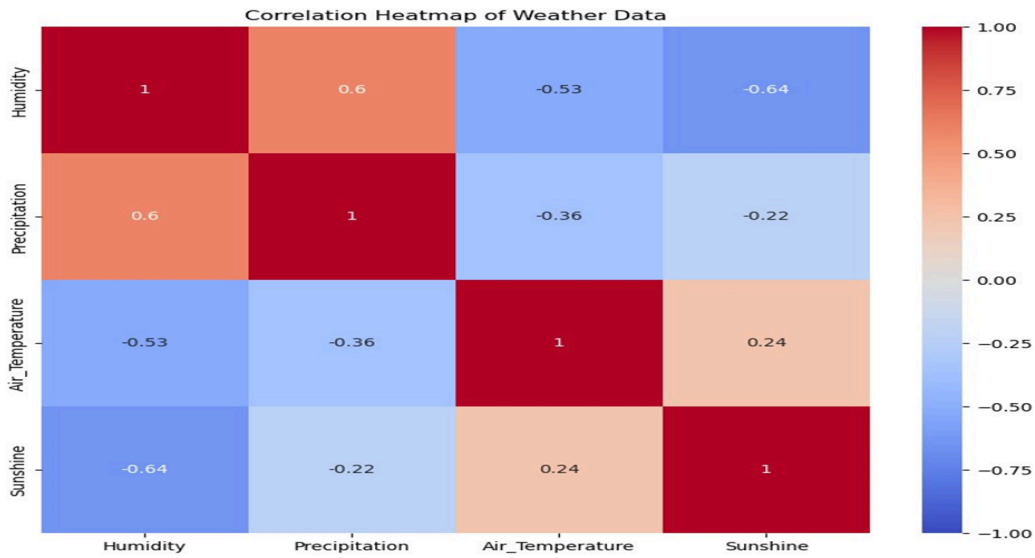
Result and Limitations

Visualisation of Data





Monthly accident data indicates a significant surge in September, following a consistent rise from January to September, possibly reflecting heightened travel activity. Conversely, a gradual decline from September to December highlights seasonal fluctuations in travel demand, underscoring the importance of adapting operational strategies and resource allocation accordingly for enhanced safety and efficiency.



A correlation heatmap of weather variables shows strong positive correlations among temperature, humidity, and sunshine, indicating they tend to increase or decrease together. Conversely, it reveals a strong negative correlation between sunshine and precipitation, suggesting that higher sunshine typically corresponds to lower precipitation.

Output Data

The final output consists of a merged dataset with monthly accident figures and corresponding weather conditions. This dataset is used to calculate correlations and generate visualisations to identify trends and patterns.

Data Structure and Quality

The output data is structured in a tabular format, with columns for month, number of accidents, temperature, precipitation, and other relevant variables. The quality is high due to thorough cleaning and preprocessing steps.

Data Format

The output is provided in CSV format, facilitating easy use in further analysis and reporting tools.

Critical Reflection

While the data quality is generally good, potential issues include the limited scope of the data (restricted to Munich) and the potential for unaccounted external factors influencing accident rates. Additionally, extreme weather events might not be fully captured by monthly averages, which could affect the analysis accuracy.

Figures and Tables

Figure 1: High-Level Data Pipeline Overview

Month	Accidents	Temperature (°C)	Precipitation (mm)
January	50	2.3	35
February	45	3.1	28
March	60	6.2	40

Table 1: Sample Merged Data

This report provides a comprehensive overview of the project, including data sources, pipeline structure, and preliminary results. The correlation between weather conditions and traffic accidents in Munich will be further analysed to draw definitive conclusions.